

Substituting the values of dp/dx and dq/dx in (ii), we get

$$\frac{d^2y}{dx^2} = -\frac{1}{q^2} \left[q \left(\frac{qr - ps}{q} \right) - p \left(\frac{qs - pt}{q} \right) \right] = -\frac{q^2r - 2pqs + p^2t}{q^3}.$$

PROBLEMS 5.5

1. If $z = u^2 + v^2$ and $u = at^2$, $v = 2at$, find dz/dt . (P.T.U., 2005)
2. If $u = \tan^{-1}(y/x)$ where $x = e^t - e^{-t}$, and $y = e^t + e^{-t}$, find du/dt . (V.T.U., 2003)
3. Find the value of $\frac{du}{dt}$ given $u = y^2 - 4ax$, $x = at^2$, $y = 2at$. (Anna, 2009)
4. At a given instant the sides of a rectangle are 4 ft. and 3 ft. respectively and they are increasing at the rate of 1.5 ft./sec. and 0.5 ft./sec. respectively, find the rate at which the area is increasing at that instant.
5. If $z = 2xy^2 - 3x^2y$ and if x increases at the rate of 2 cm. per second and it passes through the value $x = 3$ cm., show that if y is passing through the value $y = 1$ cm., y must be decreasing at the rate of $2\frac{2}{15}$ cm. per second, in order that z shall remain constant.
6. If $u = x^2 + y^2 + z^2$ and $x = e^{2t}$, $y = e^{2t} \cos 3t$, $z = e^{2t} \sin 3t$. Find $\frac{du}{dt}$ as a total derivative and verify the result by direct substitution.
7. If $\phi(cx - az, cy - bz) = 0$, show that $\frac{a\partial z}{\partial x} = \frac{b\partial z}{\partial y} = c$.
8. If $f(x, y) = 0$, $\phi(y, z) = 0$, show that $\frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial z} \cdot \frac{dz}{dx} = \frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y}$.
9. If the curves $f(x, y) = 0$ and $\phi(y, z) = 0$ touch, show that at the point of contact, $\frac{\partial f}{\partial x} \cdot \frac{\partial \phi}{\partial y} = \frac{\partial f}{\partial y} \cdot \frac{\partial \phi}{\partial x}$.
10. If $f(x, y) = 0$, show that $\left(\frac{\partial f}{\partial y}\right)^3 \frac{d^2y}{dx^2} = 2 \left(\frac{\partial f}{\partial x}\right) \left(\frac{\partial f}{\partial y}\right) \left(\frac{\partial^2 f}{\partial x \partial y}\right) - \left(\frac{\partial f}{\partial y}\right)^2 \frac{\partial^2 f}{\partial x^2} - \left(\frac{\partial f}{\partial x}\right)^2 \left(\frac{\partial^2 f}{\partial y^2}\right)$.

5.6 CHANGE OF VARIABLES

If $u = f(x, y)$... (1)

where $x = \phi(s, t)$ and $y = \Psi(s, t)$... (2)

it is often necessary to change expressions involving $u, x, y, \partial u/\partial x, \partial u/\partial y$ etc. to expressions involving $u, s, t, \partial u/\partial s, \partial u/\partial t$ etc.

The necessary formulae for the change of variables are easily obtained. If t is regarded as a constant, then x, y, u will be functions of s alone. Therefore, by (i) of page 208, we have

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial s} \quad \dots (3)$$

where the ordinary derivatives have been replaced by the partial derivatives because x, y are functions of two variables s and t .

$$\therefore \text{Similarly, regarding } s \text{ as constant, we obtain } \frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial t} \quad \dots (4)$$

On solving (3) and (4) as simultaneous equations in $\partial u/\partial x$ and $\partial u/\partial y$, we get their values in terms of $\partial u/\partial s, \partial u/\partial t, u, s, t$.

If instead of the equations (2), s and t are given in terms of x and y , say: $s = \xi(x, y)$ and $t = \eta(x, y)$, ... (5)

then it is easier to use the formulae $\frac{\partial u}{\partial x} = \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x}$... (6)

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} \quad \dots(7)$$

The higher derivatives of u can be found by repeated application of formulae (3) and (4) or of (6) and (7).

Example 5.21. If $u = F(x - y, y - z, z - x)$, prove that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0 \quad (\text{V.T.U., 2010 ; U.T.U. 2009 ; U.P.T.U., 2003})$$

Solution. Put $x - y = r, y - z = s$ and $z - x = t$, so that $u = f(r, s, t)$.

$$\begin{aligned} \therefore \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial x} \\ &= \frac{\partial u}{\partial r} \cdot (1) + \frac{\partial u}{\partial s} \cdot (0) + \frac{\partial u}{\partial t} \cdot (-1) = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial t} \end{aligned} \quad \dots(i)$$

$$\text{Similarly,} \quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial y} = -\frac{\partial u}{\partial r} + \frac{\partial u}{\partial s} \quad \dots(ii)$$

$$\text{and} \quad \frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial z} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial z} + \frac{\partial u}{\partial t} \cdot \frac{\partial t}{\partial z} = -\frac{\partial u}{\partial s} + \frac{\partial u}{\partial t} \quad \dots(iii)$$

Adding (i), (ii) and (iii), we get the required result.

Example 5.22. If $z = f(x, y)$ and $x = e^u \cos v, y = e^u \sin v$, prove that $x \frac{\partial z}{\partial v} + y \frac{\partial z}{\partial u} = e^{2u} \frac{\partial z}{\partial y}$

$$\text{and} \quad \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = e^{-2u} \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right] \quad (\text{Mumbai, 2009})$$

$$\begin{aligned} \text{Solution. We have} \quad \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} \\ &= \frac{\partial z}{\partial x} (e^u \cos v) + \frac{\partial z}{\partial y} (e^u \sin v) \end{aligned} \quad \dots(i)$$

$$\begin{aligned} \frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} \\ &= \frac{\partial z}{\partial x} (-e^u \sin v) + \frac{\partial z}{\partial y} (e^u \cos v) \end{aligned} \quad \dots(ii)$$

$$\begin{aligned} \therefore x \frac{\partial z}{\partial v} + y \frac{\partial z}{\partial u} &= (e^u \cos v) \left[-e^u \sin v \frac{\partial z}{\partial x} + e^u \cos v \frac{\partial z}{\partial y} \right] + (e^u \sin v) \left[e^u \cos v \frac{\partial z}{\partial x} + e^u \sin v \frac{\partial z}{\partial y} \right] \\ &= (e^{2u} \cos^2 v + e^{2u} \sin^2 v) \frac{\partial z}{\partial y} = e^{2u} \frac{\partial z}{\partial y} \end{aligned}$$

Now squaring (i) and (ii) and adding, we get

$$\begin{aligned} \left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 &= e^{2u} \left(\cos v \frac{\partial z}{\partial x} + \sin v \frac{\partial z}{\partial y} \right)^2 + e^{2u} \left(-\sin v \frac{\partial z}{\partial x} + \cos v \frac{\partial z}{\partial y} \right)^2 \\ \text{or} \quad e^{-2u} \left[\left(\frac{\partial z}{\partial u} \right)^2 + \left(\frac{\partial z}{\partial v} \right)^2 \right] &= \cos^2 v \left(\frac{\partial z}{\partial x} \right)^2 + \sin^2 v \left(\frac{\partial z}{\partial y} \right)^2 + 2 \sin v \cos v \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \\ &\quad + \sin^2 v \left(\frac{\partial z}{\partial x} \right)^2 + \cos^2 v \left(\frac{\partial z}{\partial y} \right)^2 - 2 \sin v \cos v \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \\ &= (\cos^2 v + \sin^2 v) \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 \end{aligned}$$

Hence
$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = e^{-2u} \left[\left(\frac{\partial z}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2 \right].$$

Example 5.23. If $x + y = 2e^\theta \cos \phi$ and $x - y = 2ie^\theta \sin \phi$, show that

$$\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \phi^2} = 4xy \frac{\partial^2 u}{\partial x \partial y}$$

(Nagpur, 2009 ; U.P.T.U., 2002)

Solution. We have $x = e^\theta (\cos \phi + i \sin \phi) = e^\theta \cdot e^{i\phi}$

[See p. 205]

$$y = e^\theta (\cos \phi - i \sin \phi) = e^\theta \cdot e^{-i\phi}$$

and

Here u is a composite function of θ and ϕ .

$$\begin{aligned} \therefore \frac{\partial u}{\partial \theta} &= \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \theta} \\ &= \frac{\partial u}{\partial x} \cdot (e^\theta \cdot e^{i\phi}) + \frac{\partial u}{\partial y} (e^\theta \cdot e^{-i\phi}) = x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \end{aligned}$$

or

$$\frac{\partial}{\partial \theta} = x \frac{\partial}{\partial u} + y \frac{\partial}{\partial y} \quad \dots(i)$$

Also

$$\frac{\partial u}{\partial \phi} = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial \phi} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial \phi} = \frac{\partial u}{\partial x} \cdot (e^\theta \cdot ie^{i\phi}) + \frac{\partial u}{\partial y} (e^\theta \cdot -ie^{-i\phi}) = ix \frac{\partial u}{\partial x} - iy \frac{\partial u}{\partial y}$$

$$\frac{\partial}{\partial \phi} = ix \frac{\partial}{\partial x} - iy \frac{\partial}{\partial y} \quad \dots(ii)$$

Using the operator (i), we have

$$\begin{aligned} \frac{\partial^2 u}{\partial \theta^2} &= \frac{\partial}{\partial \theta} \left(\frac{\partial u}{\partial \theta} \right) = \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} \right) \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \\ &= x \frac{\partial}{\partial x} \left(x \frac{\partial u}{\partial x} \right) + x \frac{\partial}{\partial x} \left(y \frac{\partial u}{\partial y} \right) + y \frac{\partial}{\partial y} \left(x \frac{\partial u}{\partial x} \right) + y \frac{\partial}{\partial y} \left(y \frac{\partial u}{\partial y} \right) \\ &= x \left(x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \right) + xy \frac{\partial^2 u}{\partial x \partial y} + yx \frac{\partial^2 u}{\partial y \partial x} + y \left(y \frac{\partial^2 u}{\partial y^2} + \frac{\partial u}{\partial y} \right) \\ &= x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \end{aligned} \quad \dots(iii)$$

Similarly using (ii),
$$\frac{\partial^2 u}{\partial \phi^2} = \frac{\partial}{\partial \phi} \left(\frac{\partial u}{\partial \phi} \right) = \left(ix \frac{\partial}{\partial x} - iy \frac{\partial}{\partial y} \right) \left(ix \frac{\partial u}{\partial x} - iy \frac{\partial u}{\partial y} \right)$$

$$= -x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} - y^2 \frac{\partial^2 u}{\partial y^2} - x \frac{\partial u}{\partial x} - y \frac{\partial u}{\partial y} \quad \dots(iv)$$

Adding (iii) and (iv), we get
$$\frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial \phi^2} = 4xy \frac{\partial^2 u}{\partial x \partial y}$$

Example 5.24. Transform the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ into polar coordinates.

(P.T.U., 2010)

Solution. We have $x = r \cos \theta$, $y = r \sin \theta$ and $r = \sqrt{(x^2 + y^2)}$, $\theta = \tan^{-1}(y/x)$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{(x^2 + y^2)}} = \cos \theta \text{ and } \frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2} = -\frac{\sin \theta}{r}$$

Thus,

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta} \cdot \frac{\partial \theta}{\partial x} = \cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta}$$

i.e., $\frac{\partial}{\partial x} = \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}$ Similarly, $\frac{\partial}{\partial y} = \sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta}$.

$$\begin{aligned} \therefore \frac{\partial^2 u}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \left(\cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\cos \theta \frac{\partial u}{\partial r} - \frac{\sin \theta}{r} \frac{\partial u}{\partial \theta} \right) \\ &= \cos^2 \theta \frac{\partial^2 u}{\partial r^2} - \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\sin^2 \theta}{r} \frac{\partial u}{\partial r} + \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} \quad \dots(i) \end{aligned}$$

and $\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \right) = \left(\sin \theta \frac{\partial}{\partial r} + \frac{\cos \theta}{r} \frac{\partial}{\partial \theta} \right) \left(\sin \theta \frac{\partial u}{\partial r} + \frac{\cos \theta}{r} \frac{\partial u}{\partial \theta} \right)$

$$= \sin^2 \theta \frac{\partial^2 u}{\partial r^2} + \frac{2 \sin \theta \cos \theta}{r} \frac{\partial^2 u}{\partial r \partial \theta} + \frac{\cos^2 \theta}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\cos^2 \theta}{r} \frac{\partial u}{\partial r} - \frac{2 \sin \theta \cos \theta}{r^2} \frac{\partial u}{\partial \theta} \quad \dots(ii)$$

Adding (i) and (ii), we get $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{1}{r} \frac{\partial u}{\partial r}$.

Hence the transformed equation is $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$.

PROBLEMS 5.6

1. If $z = f(x, y)$ and $x = e^u + e^{-v}$, $y = e^{-u} - e^v$, prove that $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y}$. (V.T.U., 2006)
2. If $u = f(r, s)$, $r = x + at$, $s = y + bt$ and x, y, t are independent variables, show that $\frac{\partial u}{\partial t} = a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y}$.
3. If $\phi(z/x^3, y/x) = 0$, prove that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 3z$. (Mumbai, 2007)
4. If $u = f(x, y)$ and $x = r \cos \theta$, $y = r \sin \theta$, prove that $\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 = \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta} \right)^2$. (V.T.U., 2010 ; Madras 2006 ; Rohtak, 2005)
5. If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$, prove that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$. (U.P.T.U., 2006 ; Raipur, 2005)
6. If $u = f(e^{y-z}, e^{z-x}, e^{x-y})$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (Mumbai, 2008 S)
7. If $u = f(r, s, t)$ and $r = x/y$, $s = y/z$, $t = z/x$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$. (Anna, 2009 ; Kurukshetra, 2006)
8. If $x = u + v + w$, $y = vw + wu + uv$, $z = uvw$ and F is a function of x, y, z , show that $u \frac{\partial F}{\partial u} + v \frac{\partial F}{\partial v} + w \frac{\partial F}{\partial w} = x \frac{\partial F}{\partial x} + 2y \frac{\partial F}{\partial y} + 3z \frac{\partial F}{\partial z}$.
9. Given that $u(x, y, z) = f(x^2 + y^2 + z^2)$ where $x = r \cos \theta \cos \phi$, $y = r \cos \theta \sin \phi$ and $z = r \sin \theta$, find $\frac{\partial u}{\partial \theta}$ and $\frac{\partial u}{\partial \phi}$.
10. If the three thermodynamic variables P, V, T are connected by a relation $f(P, V, T) = 0$, show that $\left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial T}{\partial V} \right)_P \left(\frac{\partial V}{\partial P} \right)_T = -1$.
11. If by the substitution $u = x^2 - y^2$, $v = 2xy$, $f(x, y) = \theta(u, v)$, show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 4(x^2 + y^2) \left(\frac{\partial^2 \theta}{\partial u^2} + \frac{\partial^2 \theta}{\partial v^2} \right)$. (Anna, 2003)
12. Transform $\frac{\partial^2 z}{\partial x^2} + 2xy^2 \frac{\partial z}{\partial x} + 2(y - y^3) \frac{\partial z}{\partial y} + x^2 y^2 z = 0$ by the substitution $x = uv$, $y = 1/v$. Hence show that z is the same function of u and v as of x and y .

5.7 (1) JACOBIANS

If u and v are functions of two independent variables x and y , then the determinant

$$\begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix} \text{ is called the Jacobian* of } u, v \text{ with respect to } x, y$$

and is written as $\frac{\partial(u, v)}{\partial(x, y)}$ or $J\left(\frac{u, v}{x, y}\right)$.

Similarly the Jacobian of u, v, w with respect to x, y, z is

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{vmatrix}$$

Likewise, we can define Jacobians of four or more variables. An important application of Jacobians is in connection with the change of variables in multiple integrals (§ 7.7).

(2) Properties of Jacobians. We give below two of the important properties of Jacobians. For simplicity, the properties are stated in terms of two variables only, but these are evidently true in general.

I. If $J = \partial(u, v) / \partial(x, y)$ and $J' = \partial(x, y) / \partial(u, v)$ then $JJ' = 1$.

Let $u = f(x, y)$ and $v = g(x, y)$.

Suppose, on solving for x and y , we get $x = \phi(u, v)$ and $y = \psi(u, v)$.

Then

$$\left. \begin{aligned} \frac{\partial u}{\partial u} &= 1 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial u}, \\ \frac{\partial u}{\partial v} &= 0 = \frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial v}, \\ \frac{\partial v}{\partial u} &= 0 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial u}, \\ \frac{\partial v}{\partial v} &= 1 = \frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial v}, \end{aligned} \right\} \dots(i)$$

and

$$\begin{aligned} \therefore JJ' &= \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix} \times \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial y / \partial x & \partial v / \partial y \end{vmatrix} \times \begin{vmatrix} \partial x / \partial u & \partial y / \partial u \\ \partial x / \partial v & \partial y / \partial v \end{vmatrix} \\ &\quad \text{(Interchanging rows and columns of the 2nd determinant).} \\ &= \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1. \end{aligned} \quad \text{[By virtue of (i)]}$$

II. **Chain rule for Jacobians.** If u, v are functions of r, s and r, s are functions of x, y , then

$$\begin{aligned} \frac{\partial(u, v)}{\partial(x, y)} &= \frac{\partial(u, v)}{\partial(r, s)} \cdot \frac{\partial(r, s)}{\partial(x, y)}, \\ \frac{\partial(u, v)}{\partial(r, s)} \cdot \frac{\partial(r, s)}{\partial(x, y)} &= \begin{vmatrix} \partial u / \partial r & \partial u / \partial s \\ \partial v / \partial r & \partial v / \partial s \end{vmatrix} \times \begin{vmatrix} \partial r / \partial x & \partial s / \partial x \\ \partial r / \partial y & \partial s / \partial y \end{vmatrix} \\ &\quad \text{[Interchanging rows and columns of the 2nd det.]} \end{aligned}$$

$$= \begin{vmatrix} \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial u}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial u}{\partial s} \cdot \frac{\partial s}{\partial y} \\ \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial x} & \frac{\partial v}{\partial r} \cdot \frac{\partial r}{\partial y} + \frac{\partial v}{\partial s} \cdot \frac{\partial s}{\partial y} \end{vmatrix} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y \\ \partial v / \partial x & \partial v / \partial y \end{vmatrix} = \frac{\partial(u, v)}{\partial(x, y)}.$$

* Called after the German mathematician *Carl Gustav Jacob Jacobi* (1804–1851), who made significant contributions to mechanics, partial differential equations, astronomy, elliptic functions and the calculus of variations.

Example 5.25. (i) In polar coordinates, $x = r \cos \theta$, $y = r \sin \theta$, show that

$$\frac{\partial(x, y)}{\partial(r, \theta)} = r. \quad (\text{U.P.T.U., 2006 ; V.T.U., 2004 ; Andhra, 2000})$$

(ii) In cylindrical coordinates (Fig. 8.28), $x = \rho \cos \phi$, $y = \rho \sin \phi$, $z = z$, show that

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \rho.$$

(iii) In spherical polar coordinates (Fig. 8.29), $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$, show that

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin \theta. \quad (\text{Anna, 2009 ; Hazaribagh, 2009 ; Rohtak, 2003})$$

Solution. (i) We have

$$\frac{\partial x}{\partial r} = \cos \theta, \quad \frac{\partial x}{\partial \theta} = -r \sin \theta \quad \text{and} \quad \frac{\partial y}{\partial r} = \sin \theta, \quad \frac{\partial y}{\partial \theta} = -r \cos \theta$$

$$\therefore \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

(ii) We have

$$\frac{\partial x}{\partial \rho} = \cos \phi, \quad \frac{\partial x}{\partial \phi} = -\rho \sin \phi, \quad \frac{\partial x}{\partial z} = 0,$$

$$\frac{\partial y}{\partial \rho} = \sin \phi, \quad \frac{\partial y}{\partial \phi} = \rho \cos \phi, \quad \frac{\partial y}{\partial z} = 0 \quad \text{and} \quad \frac{\partial z}{\partial \rho} = 0, \quad \frac{\partial z}{\partial \phi} = 0, \quad \frac{\partial z}{\partial z} = 1$$

$$\therefore \frac{\partial(x, y, z)}{\partial(\rho, \phi, z)} = \begin{vmatrix} \cos \phi & -\rho \sin \phi & 0 \\ \sin \phi & \rho \cos \phi & 0 \\ 0 & 0 & 1 \end{vmatrix} = \rho.$$

(iii) We have

$$\frac{\partial x}{\partial r} = \sin \theta \cos \phi, \quad \frac{\partial x}{\partial \theta} = r \cos \theta \cos \phi, \quad \frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi,$$

$$\frac{\partial y}{\partial r} = \sin \theta \sin \phi, \quad \frac{\partial y}{\partial \theta} = r \cos \theta \sin \phi, \quad \frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi,$$

and

$$\frac{\partial z}{\partial r} = \cos \theta, \quad \frac{\partial z}{\partial \theta} = -r \sin \theta, \quad \frac{\partial z}{\partial \phi} = 0.$$

$$\therefore \frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix} = r^2 \sin \theta.$$

Example 5.26. If $y_1 = \frac{x_2 x_3}{x_1}$, $y_2 = \frac{x_3 x_1}{x_2}$, $y_3 = \frac{x_1 x_2}{x_3}$, show that the Jacobian of y_1, y_2, y_3 with respect to x_1, x_2, x_3 is 4. (U.P.T.U., 2006)

Solution. We have $\frac{\partial y_1}{\partial x_1} = -\frac{x_2 x_3}{x_1^2}$, $\frac{\partial y_1}{\partial x_2} = \frac{x_3}{x_1}$, $\frac{\partial y_1}{\partial x_3} = \frac{x_2}{x_1}$

$$\frac{\partial y_2}{\partial x_1} = \frac{x_3}{x_2}, \quad \frac{\partial y_2}{\partial x_2} = -\frac{x_3 x_1}{x_2^2}, \quad \frac{\partial y_2}{\partial x_3} = \frac{x_1}{x_2}$$

and

$$\frac{\partial y_3}{\partial x_1} = \frac{x_2}{x_3}, \quad \frac{\partial y_3}{\partial x_2} = \frac{x_1}{x_3}, \quad \frac{\partial y_3}{\partial x_3} = -\frac{x_1 x_2}{x_3^2}$$

$$\therefore \frac{\partial(y_1 y_2 y_3)}{\partial(x_1 x_2 x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix} = \begin{vmatrix} -\frac{x_2 x_3}{x_1^2} & \frac{x_3}{x_1} & \frac{x_2}{x_1} \\ \frac{x_3}{x_2} & -\frac{x_3 x_1}{x_2^2} & \frac{x_1}{x_2} \\ \frac{x_2}{x_3} & \frac{x_1}{x_3} & -\frac{x_1 x_2}{x_3^2} \end{vmatrix}$$

$$= \frac{1}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -x_2 x_3 & x_3 x_1 & x_1 x_2 \\ x_2 x_3 & -x_3 x_1 & x_1 x_2 \\ x_2 x_3 & x_3 x_1 & -x_1 x_2 \end{vmatrix} = -\frac{x_1^2 x_2^2 x_3^2}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= -1(1-1) - 1(-1-1) + 1(1+1) = 0 + 2 + 2 = 4.$$

Example 5.27. In $u = x + 3y^2 - z^3$, $v = 4x^2yz$, $w = 2z^2 - xy$, evaluate $\partial(u, v, w)/\partial(x, y, z)$ at $(1, -1, 0)$.

(V.T.U., 2006)

Solution. $\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} 1 & 6y & -3z^2 \\ 8xyz & 4x^2z & 4x^2y \\ -y & -x & 4z \end{vmatrix}$

$\therefore$ At the point $(1, -1, 0)$ $\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 1 & -6 & 0 \\ 0 & 0 & -4 \\ 1 & -1 & 0 \end{vmatrix} = 4(-1+6) = 20.$

Example 5.28. If $u = x^2 - y^2$, $v = 2xy$ and $x = r \cos \theta$, $y = r \sin \theta$, find $\frac{\partial(u, v)}{\partial(r, \theta)}$.

(V.T.U., 2009 ; Madras, 2006)

Solution. We have $\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)}$

Since $u = x^2 - y^2$, $v = 2xy$

$\therefore \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2) \quad \dots(ii)$

Since $x = r \cos \theta$, $y = r \sin \theta$,

$\therefore \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r \quad \dots(iii)$

Hence, $\frac{\partial(u, v)}{\partial(r, \theta)} = 4(x^2 + y^2) \cdot r = 4(r^2 \cos^2 \theta + r^2 \sin^2 \theta) \cdot r = 4r^3$ [Using (ii) & (iii)]

(3) Jacobian of Implicit functions. If u, v, w instead of being given explicitly in terms x, y, z , be connected with them equations such as

$f_1(u, v, w, x, y, z) = 0, f_2(u, v, w, x, y, z) = 0, f_3(u, v, w, x, y, z) = 0$, then

$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} \div \frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)}$

Obs. This result can be easily generalised. It bears analogy to the result $\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y}$, where x, y are connected by the relation $f(x, y) = 0$.

Example 5.29. If $u = x, y, z$, $v = x^2 + y^2 + z^2$, $w = x + y + z$, find $\partial(x, y, z)/\partial(u, v, w)$. (U.P.T.U., 2003)

Solution. Let $f_1 = u - x, f_2 = v - x^2 - y^2 - z^2, f_3 = w - x - y - z$.

We have
$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = (-1)^3 \frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} \div \frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} \quad \dots(i)$$

Now,
$$\frac{\partial(f_1, f_2, f_3)}{\partial(x, y, z)} = \begin{vmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y & \partial f_1 / \partial z \\ \partial f_2 / \partial x & \partial f_2 / \partial y & \partial f_2 / \partial z \\ \partial f_3 / \partial x & \partial f_3 / \partial y & \partial f_3 / \partial z \end{vmatrix} = \begin{vmatrix} -yz & -xz & -xy \\ -2x & -2y & -2z \\ -1 & -1 & -1 \end{vmatrix}$$

$$= -2(x-y)(y-z)(z-x) \quad \dots(ii)$$

and
$$\frac{\partial(f_1, f_2, f_3)}{\partial(u, v, w)} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \dots(iii)$$

Substituting values from (ii) and (iii) in (i), we get

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = (-1) \times 1 / [-2(x-y)(y-z)(z-x)] = 1/2(x-y)(y-z)(z-x).$$

(4) Functional relationship. If u, u_2, u_3 be functions of x_1, x_2, x_3 then the necessary and sufficient condition for the existence of a functional relationship of the form $f(u, u_2, u_3) = 0$, is

$$J\left(\frac{u, u_2, u_3}{x_1, x_2, x_3}\right) = 0.$$

Example 5.30. If $u = x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}$, $v = \sin^{-1} x + \sin^{-1} y$, show that u, v are functionally related and find the relationship. (Kurukshetra, 2006)

Solution. We have
$$\frac{\partial u}{\partial x} = \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}}, \quad \frac{\partial u}{\partial y} = \frac{-xy}{\sqrt{(1-y^2)}} + \sqrt{(1-x^2)}$$

and
$$\frac{\partial v}{\partial x} = \frac{1}{\sqrt{(1-x^2)}}, \quad \frac{\partial v}{\partial y} = \frac{1}{\sqrt{(1-y^2)}}$$

$$\therefore \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \sqrt{(1-y^2)} - \frac{xy}{\sqrt{(1-x^2)}} & \sqrt{(1-x^2)} - \frac{xy}{\sqrt{(1-y^2)}} \\ \frac{1}{\sqrt{(1-x^2)}} & \frac{1}{\sqrt{(1-y^2)}} \end{vmatrix}$$

$$= 1 - \frac{xy}{\sqrt{[(1-x^2)(1-y^2)]}} - 1 + \frac{xy}{\sqrt{[(1-x^2)(1-y^2)]}} = 0$$

Hence u and v are functionally related i.e., they are not independent.

We have
$$v = \sin^{-1} x + \sin^{-1} y = \sin^{-1} [x\sqrt{(1-y^2)} + y\sqrt{(1-x^2)}]$$

i.e.,
$$u = \sin v$$

which is the required relationship between u and v .

PROBLEMS 5.7

1. If $x = r \cos \theta$, $y = r \sin \theta$, evaluate $\partial(r, \theta) / \partial(x, y)$, $\partial(x, y) / \partial(r, \theta)$ and prove that $\{\partial(r, \theta) / \partial(x, y)\} \cdot \{\partial(x, y) / \partial(r, \theta)\} = 1$. (V.T.U., 2010)
2. If $x = u(1-v)$, $y = uv$, prove that $JJ' = 1$. (V.T.U., 2000 S)
3. If $x = a \cosh \xi \cos \eta$, $y = a \sinh \xi \sin \eta$, show that $\partial(x, y) / \partial(\xi, \eta) = \frac{1}{2} a^2 (\cosh 2\xi - \cos 2\eta)$ (S.V.T.U., 2007)
4. If $x = e^u \sec v$, $y = e^v \tan v$, find $J = \partial(u, v) / \partial(x, y)$, $J' = \partial(x, y) / \partial(u, v)$. Hence show $JJ' = 1$. (V.T.U., 2007 S)
5. If $u = x^2 - 2y^2$, $v = 2x^2 - y^2$ where $x = r \cos \theta$, $y = r \sin \theta$, show that $\frac{\partial(u, v)}{\partial(r, \theta)} = 6r^3 \sin 2\theta$.
6. If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$. (U.T.U., 2009 ; V.T.U., 2008)

7. If $F = xu + v - y$, $G = u^2 + vy + w$, $H = zu - v + vw$, compute $\partial(F, G, H)/\partial(u, v, w)$.
8. If $u = x + y + z$, $uv = y + z$, $uvw = z$, show that $\partial(x, y, z)/\partial(u, v, w) = u^2v$.
(Kurukshetra, 2009; P.T.U., 2009 S; V.T.U., 2003)
9. If $u^3 + v^3 = x + y$ and $u^2 + v^2 = x^3 + y^3$, show that $\frac{\partial(u, v)}{\partial(x, y)} = \frac{1}{2} \frac{y^2 - x^2}{uv(u - v)}$ (U.P.T.U., 2006 MCA)
10. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$, find $\frac{\partial(u, v)}{\partial(x, y)}$. Are u and v functionally related. If so, find this relationship.
(Nagpur, 2008)
11. If $u = 3x + 2y - z$, $v = x - 2y + z$ and $w = x(x + 2y - z)$, show that they are functionally related, and find the relation.
(Nagpur, 2009)

5.8 (1) GEOMETRICAL INTERPRETATION

If $P(x, y, z)$ be the coordinates of a point referred to axes OX, OY, OZ then the equation $z = f(x, y)$ represents a surface. (Fig. 5.1)

Let a plane $y = b$ parallel to the XZ -plane pass through P cutting the surface along the curve APB given by

$$z = f(x, b).$$

As y remains equal to b and x varies then P moves along the curve APB and $\partial z/\partial x$ is the ordinary derivative of $f(x, b)$ w.r.t. x .

Hence $\partial z/\partial x$ at P is the tangent of the angle which the tangent at P to the section of the surface $z = f(x, y)$ by a plane through P parallel to the plane XOZ , makes with a line parallel to the x -axis.

Similarly, $\partial z/\partial y$ at P is the tangent of the angle which the tangent at P to the curve of intersection of the surface $z = f(x, y)$ and the plane $x = a$, makes with a line parallel to the y -axis.

(2) Tangent plane and Normal to a surface. Let $P(x, y, z)$ and $Q(x + \delta x, y + \delta y, z + \delta z)$ be two neighbouring points on the surface $F(x, y, z) = 0$. (Fig. 5.2)

Let the arc PQ be δs and the chord PQ be δc , so that (as for plane curves)

$$\lim_{Q \rightarrow P} (\delta s/\delta c) = 1.$$

The direction cosines of PQ are $\frac{\delta x}{\delta c}, \frac{\delta y}{\delta c}, \frac{\delta z}{\delta c}$ i.e., $\frac{\delta x}{\delta s} \cdot \frac{\delta s}{\delta c}, \frac{\delta y}{\delta s} \cdot \frac{\delta s}{\delta c}, \frac{\delta z}{\delta s} \cdot \frac{\delta s}{\delta c}$

When $\delta s \rightarrow 0$, $Q \rightarrow P$ and PQ tends to tangent line PT . Then noting that the coordinates of any point on arc PQ are functions of s only, the direction cosines of PT are

$$\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \quad \dots(ii)$$

Differentiating (i) with respect to s , we obtain $\frac{\partial F}{\partial x} \cdot \frac{dx}{ds} + \frac{\partial F}{\partial y} \cdot \frac{dy}{ds} + \frac{\partial F}{\partial z} \cdot \frac{dz}{ds} = 0$.

This shows that the tangent line whose direction cosines are given by (ii), is perpendicular to the line having direction ratios

$$\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \quad \dots(iii)$$

Since we can take different curves joining Q to P , we get a number of tangent lines at P and the line having direction ratios (iii) will be perpendicular to all these tangent lines at P . Thus all the tangent lines at P lie in a plane through P perpendicular to line (iii).

Hence the equation of the tangent plane to (i) at the point P is

$$\frac{\partial F}{\partial x} (X - x) + \frac{\partial F}{\partial y} (Y - y) + \frac{\partial F}{\partial z} (Z - z) = 0$$

where (X, Y, Z) are the current coordinates of any point on this tangent plane.

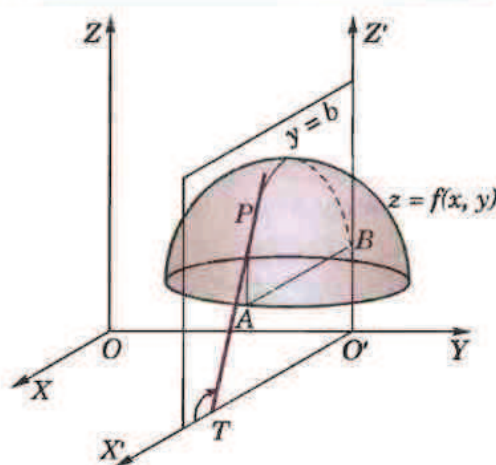


Fig. 5.1

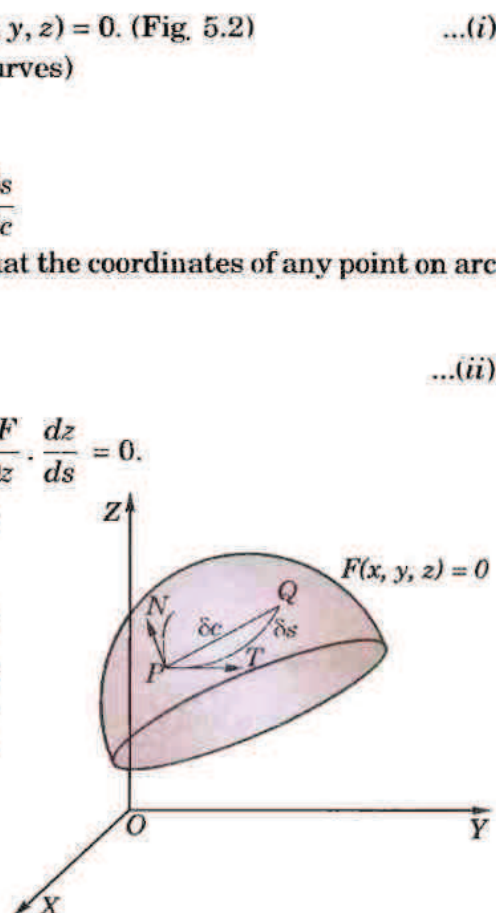


Fig. 5.2